



RAJKUMAR RAJENDRAN

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Professional Summary

- 8+ years of software development experience, specializing in Java, Angular, Spring Boot, microservices architecture with a Masters in computer science.
- Expertise in cloud platforms and containerization, including OpenShift, PCF, AWS and Docker, with notable migrations and integrations in enterprise environments.
- Designed and built an AI-based documentation agent that analyzes code diffs and generates concise, standards-compliant technical documentation, reducing manual documentation effort.
- Experience in working with relational and non-relational databases like Mongo DB, Cassandra, Postgres, Oracle Db, SQL Server, DB2 and MySQL.
- Engineered low-latency microservice workflows by optimizing thread utilization, multithreading, non-blocking I/O, and multi-layer caching, reducing response times and improving throughput under high concurrency.
- Designed and deployed cloud-native applications on Azure, leveraging Virtual Machines, Blob Storage, Functions, and Azure Front Door to build scalable and highly available services.
- Proven track record in establishing secure and efficient API connectivity, onboarding numerous endpoints, and implementing rate limits and security profiles.
- Team player with proficiency in Agile methodologies and experience in working with diverse teams.
- Created and tracked JIRA tickets for feature enhancements, production issues, and technical debt, ensuring timely resolution and cross-team visibility.
- Oracle Certified Professional Java SE 8 Programmer.

Technical Skills

Languages : Java, Java 21, Python, C++, PHP, Ruby, Typescript, SQL, JavaScript, HTML, CSS
Frameworks : Spring Boot 3.5, Rails, Flask, Laravel, Angular 19
Tools : Jenkins, Swagger, Splunk, Gemfire, Redis, Solace, Kafka, Swagger UI, Log4J, Resilience4J
Technologies : OpenShift, PCF, AWS, Microsoft Azure, Docker, Elastic search, CyberArk, Eureka server
Version Control : Git, Bitbucket, SVN
Process : Agile, Pair Programming, Test-driven Development, Scrum, Waterfall, Jira

Experience

Senior Software Developer (Technical Team Lead) at TCS, Irving, Texas.

Sep 2021 – Present

Project: CITI Bank

- Led end-to-end design, development, and migration of Spring Boot microservices from Pivotal Cloud Foundry (PCF) to Red Hat OpenShift, enabling a scalable cloud-native PaaS architecture.
- Created microservices from scratch to implement business logic and on-boarded to OpenShift/PCF containers.
- Designed latency-sensitive APIs using Spring WebClient, asynchronous processing, connection pooling, and distributed caching to minimize blocking calls and improve response times under high traffic.
- Integrated multiple RESTful service endpoints using Angular 19 HttpClient to retrieve real-time microservice data, enabling dynamic visualization of service status, response metrics, and failure indicators.
- Debugged and resolved UI defects and runtime errors by analyzing browser console logs, network traces, and Angular component behavior, improving application stability and user experience.

- Configured end-to-end connectivity for multiple external partners by whitelisting certificates and managing diverse authentication methods and role-based authorization.
- Implemented API security using OAuth 2.0 and JWT, enabling token-based authentication and authorization, secure access delegation, and stateless session management across microservices.
- Owned API platform integration by onboarding 40+ APIs to IBM API Gateway Developer Portal, configuring TLS profiles, rate limits, and quotas to ensure secure and controlled partner access.
- Designed and implemented an AI-powered Document Manager Agent that analyzes code diffs, extracts business logic changes, and auto-generates structured technical documentation, improving documentation accuracy and reducing manual effort during release cycles.
- Designed and optimized complex Oracle SQL queries, stored procedures, and indexes to improve query performance and reduce response times in high-volume transactional systems.
- Engineered the transition of mission-critical Spring Boot microservices to Azure Kubernetes Service (AKS), leveraging Azure Container Registry (ACR) and Helm charts to achieve seamless auto-scaling and high availability across availability zones.
- Implemented user session caching using Gemfire, achieving an 80% reduction in MongoDB load.
- Resolved critical production bugs in high-traffic core applications, handling over 10,000 invocations per hour.
- Served as a technical mentor and point of contact for interns and new team members, providing guidance on architecture, domain knowledge, and best development practices.

Software Developer at *Vacation Labs, Goa, India.*

May 2018 – Aug 2019

Project: Unified Booking Management and Tour Operator System

- Developed back-office application and microservices to store booking using Java Spring Boot.
- Optimized Docker container for staging deployments, reducing overall time to execute by 60%
- Performed SQL (Oracle SQL database) performance tuning using execution plans, indexing strategies, and query rewrites, reducing critical report generation time by over 70%.
- Debugged API endpoints and live logs to solve production errors.
- Participated in code reviews and pull request workflows, ensuring code quality and adherence to team standards.
- Integrated and maintained pull requests from the team using Git.
- Assisted in designing service contracts and request/response models for microservice interactions.
- Hosted containerized microservices on AWS EC2 instances running Kubernetes (Amazon EKS), enabling flexible scaling and efficient resource utilization.

Software Developer at *NTT Data, Bangalore, India.*

Aug 2016 – Apr 2018

Project: BMW

- Added Java programs to map inputs from Frontend to Legacy systems through IIB nodes.
- Partnered with Product, QA, and DevOps teams to align sprint goals with business priorities, ensuring predictable delivery of high-availability financial services.
- Converted monolithic legacy application to microservices using Java Spring Boot.
- Worked extensively with PL/SQL procedures, triggers, and bulk operations to handle large-scale financial transaction processing.
- Designed low-latency content delivery architecture using S3 for static storage and CloudFront for edge caching.
- Achieved loose coupling through Service Oriented Architectural (SOA) pattern.
- Addressed regular production bugs and improvements in existing applications that increased audience reach.
- Reduced reporting API endpoint response times from several minutes to fewer than 10 seconds by rewriting the API, optimizing SQL queries, and adding a caching layer using Redis.

- Deployed code to push logs to Paper trail and created instant email alert for production errors.

Education

Doctorate in Information Studies - Trine University	Feb 2026 – In Progress
Masters in computer science - University of Texas at Arlington	Aug 2019 – May 2021
Bachelor of Engineering in Electrical and Electronics - Anna University	Aug 2012 – May 2016

Certification

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| • Oracle Certified Professional Java SE 8 Programmer | Oct 2021 |
| • Microsoft Certified: Azure Developer Associate | Aug 2025 |

Individual Projects

Job Analysis (Natural Language processing): Trained Google's word2vec neural network model to learn associations between various jobs and relevant skills using data collected by scrapping job listing websites.

Car rental Android application: Designed and developed a scalable Android application for renting cars using MVC design pattern with three different user roles.

Restaurant inventory management: Created a responsive web application for restaurant business using PHP Laravel framework and MySQL database with admin controls.